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Development of a microstructural grand potential-based sintering model

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ABSTRACT

Microstructure is a controlling factor in the behavior of sintered materials. This work presents a quantitative phase field model of thermal sintering that predicts the evolution of the microstructure by capturing the sintering stress, GB/vacancy interactions, non-uniform diffusion, and grain coarsening without introducing a separate rigid body motion term. The model provides a mechanistic description of sintering using the grand potential phase field approach. Small test simulations are used to verify the new model against sintering theory, and they show that 3D simulations predict faster densification and coarsening than 2D simulations. 3D simulations are compared against experimental data available in the literature. The results of this comparison show that the model provides a reasonable estimate of the sintering behavior, though it overpredicts the sintering rate. This may be due to uncertainty in the material parameters and the relatively small scale of the simulation.

1. Introduction

Bulk material properties, such as thermal conductivity, strength, and resistance to fracture are largely determined by the material microstructure. In sintered materials, the microstructure is determined by the sintering conditions [1]. Thermal sintering is a common manufacturing process in metals and ceramics in which a highly-porous powder compact (the green body) is heated so that the particles coalesce into a dense, polycrystalline solid without melting. Sintering in crystalline solids occurs due to vacancy diffusion, which causes the total surface area to decrease by forming grain boundaries (GBs) [2]. The material densifies as vacancies diffuse along GBs from interior pores to external surfaces [3,4]. In addition, the GBs migrate to reduce the total GB area in a process known as coarsening; this GB migration is slowed by interactions with pores [5,6]. The sintering process is shown schematically in Fig. 1.

Determining the proper sintering conditions to produce desired properties requires the ability to quantitatively predict microstructure evolution during the sintering process. There are four governing mechanisms which must be accounted for in any quantitative microstructural sintering model:

- Sintering stress—The driving force for densification during sintering is the energy difference between surfaces and GBs. As a GB is formed at the neck of two particles, it reduces the free surface area while increasing the GB area, thus reducing the overall surface energy and increasing the overall GB energy [6]. Only when the surface energy is larger than one half the GB energy is there a thermodynamic driving force for sintering—since two surfaces are eliminated in the formation of a single GB. The energy difference between surfaces and GBs is known as the sintering stress. The surface and GB energies interact when GBs intersect surfaces, resulting in a contact (or dihedral) angle θ, shown in Fig. 1, that is controlled by the ratio of the GB to surface energy.
- GB/vacancy interaction—GBs play a major role in the net elimination of vacancies that results in densification. The role of GBs during sintering can be conceptualized in two ways, both of which can explain densification. The first is that GBs act as sinks of vacancies and the second is that GBs serve as rapid diffusion paths for vacancies to reach free surfaces.
- Non-uniform diffusion—The kinetics of densification during sintering are governed by the rate at which vacancies diffuse through the material. However, diffusion does not occur at a uniform rate throughout the material. It is accelerated on GBs and surfaces because there is more empty space for atomic vibrations. Thus, GB and surface diffusion are rate-controlling factors for sintering kinetics [7].
- Grain coarsening—The driving force for coarsening is the reduction in the total grain boundary energy as large grains grow and small grains shrink. The kinetics are dictated by the GB mobility and by the interactions with pores.

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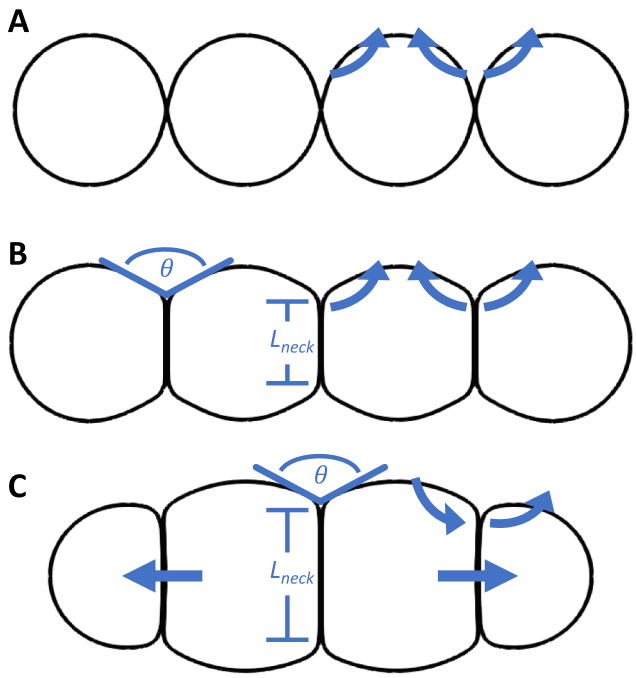


Fig. 1. Simple demonstration of sintering theory. A: Four particles inside a powder compact. Vacancies diffuse from contact regions between particles to free surfaces, increasing the particle contact area and changing particle shape. Arrows show the direction of vacancy motion. B: Particle contact regions grow into necks of width L_{neck} to form GBs. There is a material-specific dihedral angle, θ, observed where GBs meet surfaces. C: Necks fully form into GBs. The neck width and dihedral angle are also shown. GB motion consumes small grains at the expense of large ones. Additional arrows show the direction of GB motion. The dihedral angle remains constant throughout grain growth.

Other mechanisms may also play a role in thermal sintering in certain situations, such as Ostwald ripening, grain boundary sliding, evaporation/condensation, and charged defects (in ceramics). These mechanisms are not included in the model presented in this work, but may be in the future as required for other materials by using a more generalized form of the model [8].

There are several models available to describe sintering. Macroscale descriptions, such as the master sintering curve, predict the overall density and porosity change of the green body, but do not predict the evolution of the microstructure, as summarized in [6]. There are also analytical models based entirely on sintering theory [9] that exist for specific idealized cases involving only a few particles. They cannot predict the density change of the green body, but they are useful for describing sintering mechanics and verifying more advanced models. Microstructural sintering models resolve the individual particles of a green body and predict both the densification and microstructure evolution. Some of these include the rigid body motion sintering model [7], the surface evolver model [10], the discrete element method [11], and the Monte Carlo Potts model [12–15]. Due to the large computational cost of resolving individual particles, microstructural models are not typically used to model an entire green body.

Another approach that has emerged as a powerful means of creating microstructure sintering models is the phase field method. The phase field method represents the microstructure using continuous variable fields [16,17] that evolve according to partial differential equations to minimize the free energy. Interfaces between microstructural features are represented by regions where variables smoothly transition between equilibrium values. The first attempt to apply the phase field method to modeling sintering [18] suggested that rigid body motion could be used to account for densification due to vacancy annihilation at GBs. Wang [7] formulated such a rigid body motion model for general use in phase field simulations. He also defined different isotropic diffusion rates in the bulk, on GBs, and on surfaces. Other early attempts did not consider any interactions between GBs and vacancies [19,20]. Later work relied only on accelerated isotropic diffusion rates to consider the interactions between GBs and vacancies [21,22]. Deng [23] presented a model that treated GBs as rapid vacancy diffusion paths to free surfaces by defining accelerated anisotropic diffusion at GBs only parallel to the GB, and added a similar anisotropic diffusion term along surfaces. Others used a similar model to look at pore/GB interactions during the later coarsening stage of sintering [24–28] and the sintering of silver nanoparticles [29]. The rigid body motion model continued to be used in other work [30], and it was later coupled with the discrete element method [31], and improved to better consider rotations [32]. It was also coupled with solid mechanics and heat conduction [32], anisotropic GB energy [33] and mechanical contact [34]. Hötzer et al. [35] used a grand potential form to create a more quantitative model that used accelerated isotropic diffusion at GBs and surfaces. More specific models have also been created for multiple types of powders [36] and liquid phase and viscous sintering [37–40]. Of all these models, only two have been used to model more than two particles in 3D [22,35], none have used simulated particles of various sizes, and none have compared directly to experimental data.

In this work, a new quantitative and material specific grand potential phase field model for sintering is presented. It provides complete control over the surface and grain boundary energies to properly describe the sintering stress. It describes densification considering anisotropic transport of vacancies along GBs and surfaces, and it accurately captures grain coarsening. It builds on many recent developments in phase field modeling [41,24,5,42], including the grand potential model [43,8,35], to accurately describe densification rates without rigid body motion physics. The model is derived is Section 2. Then, it is verified against analytical expressions and the thermodynamic and kinetic behavior is investigated in Section 3. In Section 4, initial comparisons against sintering data for UO_2 green bodies are carried out.

Note that the term “qualitative” is used in this work to describe a phase field model that predicts microstructure evolution of a generic material with non-dimensional values. Its accuracy is evaluated by comparing distinguishing features of the microstructure with those from experimental micrographs; the accuracy of the comparison depends on the accuracy of the mechanisms included in the model. The term “quantitative” is used to describe a phase field model that predicts microstructure evolution of a specific material with actual dimensions of time, space, and energy. Its accuracy is evaluated by directly comparing measurable values at specific times with experimental data; the accuracy of the comparison depends on the accuracy of the mechanisms but also on accuracy of the material-specific parameters.

2. Model description

This grand potential sintering model describes microstructure evolution during sintering, including densification and coarsening, for a specific material and with actual dimensions of space, time, and energy. In the model, the surface and GB energies for a specific material are defined, which quantifies the sintering stress. The model assumes vacancies segregate to GBs and then rapidly diffuse along them to surfaces. The diffusion coefficients for bulk, GB and surface diffusion are defined, and GB and surface diffusion occur only along their interfaces. Finally, the GB mobility is defined, and its value, along with void interactions, control the coarsening kinetics.

A grand potential phase field model is used to provide an efficient means of allowing the interfacial energies to be defined independently of the chemical free energy of the system. The grand potential formulation used in this work is based on the model developed by Aagesen et al. [8], which relies on the model proposed by Plapp [43] and the thermodynamic analysis of Moelans [41]. A different grand potential formulation was also proposed by Choudhury et al. [44] (with recent

application to sintering by Hötzer et al. [35]), but their derivation takes a different form and is not used in this work. Other models, like the Kim-Kim-Suzuki model [45], could also be used to independently define the interfacial and chemical energies. A possible alternative approach would be to use analytical expressions to separate the interfacial and chemical energies.

This section provides the mathematical basis for the model. In Section 2.1, the governing equations for the model are described. In Section 2.2, the assignment of parameter values to make a quantitative sintering model is described. Finally, in Section 2.3, values are assigned to properties in order to model the well-studied material UO_2.

2.1. Model formulation

There are two regions considered in the model. The solid region is represented by the subscript s and the void region is represented by the subscript v. There are n order parameter variables associated with the solid region, $\vec{\eta} = \{\eta_1, \eta_2, ..., \eta_n\}$, that distinguish the various grains. There is one order parameter variable associated with the void region, ϕ, which represents all pores and the external void region. The order parameter values are constant within a region, but smoothly transition between values across interfaces. The two regions of the domain and the interfaces between them are defined by these variables as follows:

- A grain is where $\eta_i = 1$, $\eta_j = 0$ for all $j \neq i$, and $\phi = 0$.
- A void is where $\phi = 1$ and $\vec{\eta} = 0$.
- A GB is where $0 < \eta_i < 1$, $0 < \eta_j < 1$ for some $j \neq i$, and $\phi = 0$.
- A surface is where $0 < \eta_i < 1$, $\eta_j = 0$ for all $j \neq i$, and $0 < \phi < 1$.

To represent the fraction of lattice sites occupied by vacancies, which are subject to conservation of mass, the conserved variable field c is used.

The evolution equations are derived from a functional defining the total grand potential Ω according to

$$\Omega = \int_V [\omega_b(\vec{\eta}, \phi) + \omega_{gr}(\nabla\vec{\eta}, \nabla\phi) + h_s(\phi)\omega_s + h_v(\phi)\omega_v]dV, \tag{1}$$

where ω_s and ω_v are the grand potential densities for the solid and void phases, respectively, and ω_b and ω_{gr} are the bulk and gradient contributions, respectively. h_s and are switching functions that interpolate smoothly between values corresponding to the two regions and have the constraint $h_s + h_v = 1$. The bulk contribution to the grand potential density ensures that the order parameters have the correct values in the different regions and is calculated according to the equation [41]

$$f_b(\vec{\eta}, \phi) = \epsilon(\phi)\left[\left(\frac{\phi^4}{4} - \frac{\phi^2}{2}\right) + \sum_{i=1}^{n}\left(\frac{\eta_i^4}{4} - \frac{\eta_i^2}{2}\right) + \gamma\left(\phi^2\sum_{i=1}^{n}\eta_i^2 + \sum_{i=1}^{n}\sum_{j>i}^{n}\eta_i^2\eta_j^2\right) + \frac{1}{4}\right], \tag{2}$$

where ϵ is the free energy barrier coefficient and γ is the interface profile coefficient; in this work, $\gamma = 1.5$ to provide symmetric interfaces [41]. The gradient energy is calculated as [16,17]

$$f_{gr}(\nabla\vec{\eta}, \nabla\phi) = \frac{\kappa(\phi)}{2}\left[|\nabla\phi|^2 + \sum_{i=1}^{n}|\nabla\eta_i|^2\right], \tag{3}$$

where κ is the gradient energy coefficient. The grand potential densities of the two regions are defined as

$$\begin{aligned}\omega_s &= f_s - c_s\tfrac{\mu}{V_a}\\ \omega_v &= f_v - c_v\tfrac{\mu}{V_a},\end{aligned} \tag{4}$$

where f_s and f_v are the Helmhotz free energy densities of the solid and void regions, respectively, V_a is the atomic volume of the material, μ is the chemical potential of the vacancies, and c_s and c_v are the vacancy concentrations in the solid and void regions, such that $c = h_s c_s + h_v c_v$.

The Helmhotz energies of the solid and void regions are assumed to be parabolic [8]:

$$\begin{aligned}f_s(c) &= \tfrac{1}{2}k_s(c - c_s^{eq})^2,\\ f_v(c) &= \tfrac{1}{2}k_v(c - c_v^{eq})^2,\end{aligned} \tag{5}$$

where c_s^{eq} and c_v^{eq} are the equilibrium vacancy concentrations in the two regions and k_s and k_v are parabolic energy coefficients. This assumption is used to simplify the numerical solution of the problem and it is fairly accurate as long as c_s and c_v stay near their equilibrium values, which is true for this model even across the interfaces. Dilute and ideal solution models could also be used for the energies [43], but they hurt the numerical convergence. The chemical potential is derived from the Hemholtz free energies as

$$\begin{aligned}\mu &= V_a\tfrac{\partial f_s(c_s)}{\partial c_s} = V_a\tfrac{\partial f_v(c_v)}{\partial c_v},\\ &= V_a k_s(c_s - c_s^{eq}) = V_a k_v(c_v - c_v^{eq}).\end{aligned} \tag{6}$$

Eq. (6) is inverted to give

$$\begin{aligned}c_s &= \tfrac{\mu}{V_a k_s} + c_s^{eq},\\ c_v &= \tfrac{\mu}{V_a k_v} + c_v^{eq},\end{aligned} \tag{7}$$

$$c = h_s\left(\frac{\mu}{V_a k_s} + c_s^{eq}\right) + h_v\left(\frac{\mu}{V_a k_v} + c_v^{eq}\right). \tag{8}$$

Eq. (7) is substituted into Eq. (4) to obtain

$$\begin{aligned}\omega_s(\mu) &= -\tfrac{1}{2}\tfrac{\mu^2}{V_a^2 k_s} - c_s^{eq}\tfrac{\mu}{V_a},\\ \omega_v(\mu) &= -\tfrac{1}{2}\tfrac{\mu^2}{V_a^2 k_v} - c_v^{eq}\tfrac{\mu}{V_a}.\end{aligned} \tag{9}$$

The order parameters are evolved with time using modified Allen Cahn equations:

$$\begin{aligned}\tfrac{\partial\eta_i}{\partial t} &= -L_s\tfrac{\delta\Omega}{\delta\eta_i},\\ \tfrac{\partial\phi}{\partial t} &= -L_v\tfrac{\delta\Omega}{\delta\phi},\end{aligned} \tag{10}$$

where L_s and L_v are the order parameter mobilities. In this work the model is simplified by making both mobilities constant, such that the GB and surface mobilities are isotropic; anisotropic mobilities could be considered in future work using an approach similar to that developed by Moelans et al. [41]. In the grand potential formation, the chemical potential μ is treated as an independent variable and the total concentration c is calculated from μ using Eq. (8). The chemical potential is evolved with time according to

$$\frac{\partial\mu}{\partial t} = \frac{1}{\chi}\left[\nabla\cdot(\chi\mathbf{D}\nabla\mu) - \frac{1}{V_a}\frac{\partial c}{\partial\phi}\frac{\partial\phi}{\partial t}\right], \tag{11}$$

where $\mathbf{D}$ is the diffusivity tensor and χ is the susceptibility, such that

$$\chi = \frac{1}{V_a}\frac{\partial c}{\partial\mu} = \frac{1}{V_a^2}\left(\frac{h_s}{k_s} + \frac{h_v}{k_v}\right). \tag{12}$$

2.2. Quantitative parameter assignment

To make the grand potential model quantitative and material specific, it is necessary to go into more detail about the assignment of individual parameter values. These values and their partial derivatives were implemented in the implicit finite element code MARMOT [46]. Not only do partial derivatives show up in the governing equations, such as Eqs. (10) and (11), but additional derivatives are required to derive the Jacobian matrix (a matrix of partial derivatives needed for the solution of the nonlinear system created by the implicit solution).

However, for the sake of brevity, calculations of the partial derivatives are left to the reader.

First, the free energy barrier coefficient ϵ from Eq. (2) and the gradient energy coefficient κ from Eq. (3) are defined as

$$\begin{aligned}\epsilon(\phi) &= h_S(\phi)\epsilon_S + h_{GB}(\phi)\epsilon_{GB},\\ \kappa(\phi) &= h_S(\phi)\kappa_S + h_{GB}(\phi)\kappa_{GB},\end{aligned} \tag{13}$$

where h_S and h_{GB} are functions that switch between surface and grain-boundary interfaces, and ϵ_S, ϵ_{GB}, κ_S, and κ_{GB} are corresponding constant values on those interfaces. The values for ϵ_S, ϵ_{GB}, κ_S, and κ_{GB} can be defined in terms of the width of the diffuse interface l, the isotropic GB energy σ_{GB} and the isotropic surface energy σ_S following the thermodynamically consistent approach from Moelans [41]:

$$\begin{aligned}\epsilon_S &= 6\tfrac{\sigma_S}{\ell},\ \kappa_S = \tfrac{3}{4}\sigma_S\ell,\\ \epsilon_{GB} &= 6\tfrac{\sigma_{GB}}{\ell},\ \kappa_{GB} = \tfrac{3}{4}\sigma_{GB}\ell.\end{aligned} \tag{14}$$

The partial derivatives of ϵ and κ with respect to ϕ must be considered in the evaluation of Eq. (10).

There are now four switching functions (h_s, h_v, h_S, h_{GB}) that are needed, along with their partial derivatives, to solve Eqs. (10) and (11). h_s and differentiate between the solid and void regions, and therefore are functions of ϕ. h_S and h_{GB} differentiate between surfaces and GBs, which can also be done using only ϕ: interfaces with $\phi > 0$ are surfaces and those with $\phi = 0$ are GBs.For simplicity, the four switching functions are based on the same second-order continuous piecewise polynomial:

$$H(\phi, \phi_0) = \begin{cases} 0, & \phi \leqslant 0 \\ 6\left(\frac{\phi}{\phi_0}\right)^5 - 15\left(\frac{\phi}{\phi_0}\right)^4 + 10\left(\frac{\phi}{\phi_o}\right)^3, & 0 < \phi < \phi_0, \\ 1, & \phi \geqslant \phi_0 \end{cases} \tag{15}$$

where $0 < \phi_0 \leqslant 1$. The location of the transition from zero to one moves closer to $\phi = 0$ as the value of ϕ_0 decreases, as illustrated in Fig. 2. The piecewise components of Eq. (15) ensure that the switching functions are limited to the domain [0, 1] even in the case of numerical noise causing ϕ to leave these limits. h_s and are based on Eq. (15) with $\phi_0 = 1$:

$$h_v = H(\phi, 1), \quad h_s = 1 - H(\phi, 1). \tag{16}$$

h_S and h_{GB} use Eq. (15) with a user-chosen constant value of ϕ_0 that should be less than one:

$$h_S = H(\phi, \phi_0), \quad h_{GB} = 1 - H(\phi, \phi_0). \tag{17}$$

The value of ϕ_0 for h_S and h_{GB} is selected to provide the optimal balance between accuracy and numerical performance. The correct surface properties should be applied to the entire surface, suggesting a value of ϕ_0 near zero, such that $h_S = 1$ would occur for ϕ close to zero (see Fig. 2) and nearly all of the surface would have the correct properties. However, this behavior would result in very sharp transitions that would hurt numerical performance. The best numerical performance would occur for $\phi_0 = 1$, which gives the most gradual transitions. However, this behavior would result in $h_S = 1$ occurring for ϕ close to one and only a small amount of the surface would have the correct properties. Thus, it is necessary to explore the impact of ϕ_0 on the accuracy and numerical performance to determine the optimal value of ϕ_0. This is done in Section 2.3.

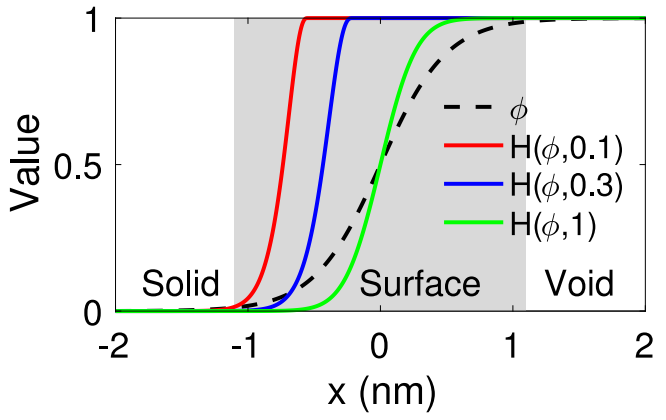


Fig. 2. Illustration of the value of the function $H(\phi, \phi_0)$, defined in Eq. (15), across a surface centered at $x = 0$ with interface width of 2 nm. Curves are shown for $\phi = 0.1$, 0.3 and 1.0. The solid and void regions are indicated on the plot and the surface is shown in gray.

The diffusivity tensor is defined as [24]:

$$\mathbf{D} = \mathbf{D}_B + \mathbf{D}_{GB} + \mathbf{D}_S, \tag{18}$$

$$\mathbf{D}_B = D_B\mathbf{I}, \tag{19}$$

$$\mathbf{D}_{GB} = D_{GB}\sum_{i=1}^{n}\sum_{j\neq i}\eta_i\eta_j\mathbf{T}_{GB}^{ij}, \tag{20}$$

$$\mathbf{D}_S = D_S\phi^2(1-\phi)^2\mathbf{T}_S, \tag{21}$$

$$\mathbf{T}_{GB}^{ij} = \mathbf{I} - \frac{\nabla\eta_i - \nabla\eta_j}{|\nabla\eta_i - \nabla\eta_j|} \otimes \frac{\nabla\eta_i - \nabla\eta_j}{|\nabla\eta_i - \nabla\eta_j|}, \tag{22}$$

$$\mathbf{T}_S = \mathbf{I} - \frac{\nabla\phi}{|\nabla\phi|} \otimes \frac{\nabla\phi}{|\nabla\phi|}. \tag{23}$$

$\mathbf{D}_B$, $\mathbf{D}_{GB}$, and $\mathbf{D}_S$ are the diffusivity tensors for the bulk, GB, and surface regions, respectively. D_B, D_{GB}, and D_S are the diffusivity magnitudes. $\mathbf{I}$ is the identity tensor, and $\mathbf{T}_{GB}^{ij}$ and $\mathbf{T}_S$ are the GB and surface directional tensors. The form of $\mathbf{D}_B$ assumes isotropic bulk diffusion, and the forms of $\mathbf{D}_{GB}$ and $\mathbf{D}_S$ only allow diffusion along GBs and surfaces, respectively. If known, D_B, D_{GB}, and D_S can be used directly. If not, then D_{GB} and D_S can be approximated by adding weights to the bulk diffusivity according to:

$$\begin{aligned}D_{GB} &= w_{GB}D_B,\\ D_S &= w_S D_B,\end{aligned} \tag{24}$$

where D_B is calculated using the standard Arrhenius equation:

$$D_B = D_0\exp\left(-\frac{E_m^B}{k_B T}\right). \tag{25}$$

D_0 is the diffusivity prefactor, E_m^B is the migration energy of vacancies in the bulk, k_B is Boltzmann's constant, and T is the absolute temperature.

The solid phase order parameter mobility is a function of the solid phase (GB) mobility, M_s, and the interface width according to:

$$L_s = \frac{4}{3}\frac{M_s}{\ell}. \tag{26}$$

The GB mobility is defined as:

$$M_s = M_0\exp\left(-\frac{Q}{k_B T}\right), \tag{27}$$

where M_0 is the mobility prefactor and Q is the mobility activation energy.

For the void-surface interface, it is assumed that the motion of the interface is limited by the diffusion of vacancies, and not by the mobility of the interface itself since atoms at the solid-void interface can rearrange relatively easily. To ensure that this is the case, the phase field void mobility is set to

$$L_v = mL_s, \tag{28}$$

such that L_v is sufficiently large to maintain diffusion-controlled behavior. The value for m must be determined by looking at the behavior at all temperatures at which the model will be applied. A similar approach would be to make it a function of the diffusion coefficient, but due to the use of a heterogeneous diffusion tensor, that approach would

require additional assumptions.

The equilibrium concentrations from Eq. (4) also need to be determined. Vacancies are unoccupied sites in the crystal lattice. In classical sintering theory [9], they are treated as a solute that can precipitate out into a second phase, pores, that have a molar fraction of one. This model follows the same reasoning, such that the equilibrium vacancy concentration within void regions is $c_v^{eq} = 1$. External open space is also treated as void region. For the solid phase, more behavior must be accounted for. Vacancies segregate to GBs and move to the external surfaces via GB diffusion. In order to account for this, the solid phase vacancy concentration is defined as

$$\begin{aligned} c_s^{eq} &= c_B^{eq} + 4(c_{GB}^{eq} - c_B^{eq})(1-\lambda)^2, \quad \text{where} \\ \lambda &= \sum_{i=1}^{n} \eta_i^2 \\ c_B^{eq} &= \exp\left(-\frac{E_f}{k_B T}\right). \end{aligned} \tag{29}$$

Here, the value of the vacancy concentration transitions between its bulk value c_B^{eq} and its GB value c_{GB}^{eq}. λ is an interpolation parameter that is equal to one in grains and zero at the center of a GB. E_f is the vacancy formation energy in the bulk. Thus, the value and partial derivatives of c_s^{eq} must be considered when evolving the order parameters.

This grand potential sintering model includes all of the governing mechanisms listed in Section 1. It includes the sintering stress by separately defining the surface and GB energies. Vacancy segregation to GBs is included using the solid phase vacancy concentration c_s^{eq}. Non-uniform diffusion is accounted for through the diffusivity tensor $\mathbf{D}$, which distinguishes between bulk, surface, and GB diffusion both in terms of direction and magnitude. Thus, the vacancies can diffuse along GBs and surfaces faster than through the bulk. Grain coarsening is a well-established feature of the phase field method [16,17] and has also been observed in the grand potential formulation this work is based on [8].

2.3. Material properties

In order to compare this model to experimental data, the material properties must be chosen to match a real material. For this work, the material uranium dioxide (UO_2) is used. The vacancy concentration describes U vacancies, since U-lattice vacancies are much less mobile than the O-lattice vacancies and limit the overall diffusion rate [47]. Many of the UO_2 vacancy property values can be taken directly from the literature. All of the parameters used in the model, including these physical parameters and the few model parameters are shown in Table 1.

According to Hall et al. [48], the GB and surface energies are difficult to measure experimentally and depend heavily on the experimental setup. Therefore, the experimental values from the literature have significant scatter. However, the ratio of the average surface to GB energy remains relatively constant at $\sigma_S/\sigma_{GB} = 2$ no matter the experimental setup. For this reason, it was decided to use a GB energy σ_{GB} value obtained via atomistic simulation [49] for a $\Sigma 5$ tilt boundary and define the isotropic surface energy $\sigma_S = 2\sigma_{GB}$. Since the GB energy value was only calculated at a single temperature, any temperature dependence in the GB energy is neglected.

The diffusivities are calculated according to Eqs. (24) and (25). However, the bulk, GB, and surface diffusivities are also subject to an extremely large experimental error. Various values from experiments and atomistic simulations from the literature are shown in Fig. 3. At the lowest temperature shown, the bulk diffusivities vary by a factor of 10^{15}. Matzke et al. [47] found in 1987 that experiments prior to 1975 had a systemic error causing their results to be several orders of magnitude too large. But even discounting those values, the bulk diffusivities still vary by up to a factor of 10^{10}. Therefore, the value used in the sintering model was calculated based on the three diffusivities taken since 1987 [47,55,56] by using the average bulk migration energy (E_m^B) and the log-average diffusivity prefactor (D_0). The values used are $E_m^B = 3.61$ eV and $D_0 = 8.33 \times 10^9$ nm^2/s. The GB weight was chosen to be $w_{GB} = 10^6$ in order to keep it approximately average between two GB diffusivities calculated since 1975 [55,57]. A literature review did not find post-1975 values for the surface diffusion, though some insights can be gained by comparing measured surface and GB diffusion rates from the pre-1975 values. The surface diffusivities were roughly eight orders of magnitude larger than the GB diffusivities. However, convergence problems began to manifest in the model when the surface diffusion weight was too large. Therefore, the surface diffusion weight was chosen to be a large enough to be orders of magnitude larger than the GB diffusion weight, but not large enough to cause convergence problems: $w_S = 10^9$.

Table 1
Physical and model parameters used in this model to represent UO_2, including where the values were found in the literature, where applicable. The justification for the selection of the ϕ_0 value is shown in Fig. 4.

Parameter	Value	Units	Ref.
	Physical parameters		
V_a	0.04092	nm^3	[50]
σ_{GB}	9.86	eV/nm^2	[49]
σ_S	$2\sigma_{GB}$	eV/nm^2	[48]
M_0	1.476×10^9	nm^4/(eV s)	[51]
Q	2.77	eV	[51]
E_f	2.69	eV	[52]
k_s	Eq. (31)	eV/nm^3	[52]
c_{GB}^{eq}	0.189		[53,54]
D_0	8.33×10^9	nm^2/s	[47,55,56]
E_m^B	3.61	eV	[47,55,56]
w_{GB}	10^6		[55,57]
w_S	10^9		[58–60]
	Model parameters		
m	40		
k_v	$10k_s$	eV/nm^3	
ϕ_0	0.3		

The phase mobility multiplication factor was chosen to be $m = 40$ because it was the lowest value at which the void phase mobility did not significantly affect the simulation results at any of the temperatures used in this work.

The ideal approach for determining a value for the increased equilibrium vacancy concentration on GBs c_{GB}^{eq} would be to use U vacancy formation and segregation energies on UO_2 grain boundaries from atomistic calculations; however, a literature search did not locate such calculations. Therefore, to determine the value of c_{GB}^{eq}, the free volume at GBs was assumed to be caused by U-vacancies. Brutzel and Vincent-Aublant found that GBs take up 15% more volume than the same number of atoms in the bulk [53] and Freyss et al. found that vacancies are 31% smaller than U atoms [54]. From these findings, the GB equilibrium concentration is determined to be

$$c_{GB}^{eq} = x(1 - c_U^{GB}) = \frac{V_U}{V_v}\left(1 - \frac{V_B}{V_{GB}}\right) = 0.189, \tag{30}$$

where x is the number density ratio of U vacancies to U atoms and c_U^{GB} is the concentration of U atoms on GBs.

The value for parabolic free energy coefficient k_s was determined by fitting a parabolic curve to the ideal solution model with a vacancy formation energy obtained from first principles calculations [52] from $c = 0$ to $c = 1$ at various sintering temperatures. The parabolas were then fit to a linear temperature equation:

$$k_s \left\langle \frac{eV}{nm^3} \right\rangle = 157.16 - 0.0025T\langle K \rangle. \tag{31}$$

Since voids do not have a measurable energy, k_v is not an easily obtained value. So it is assumed that $k_v = 10k_s$ in order to ensure that the

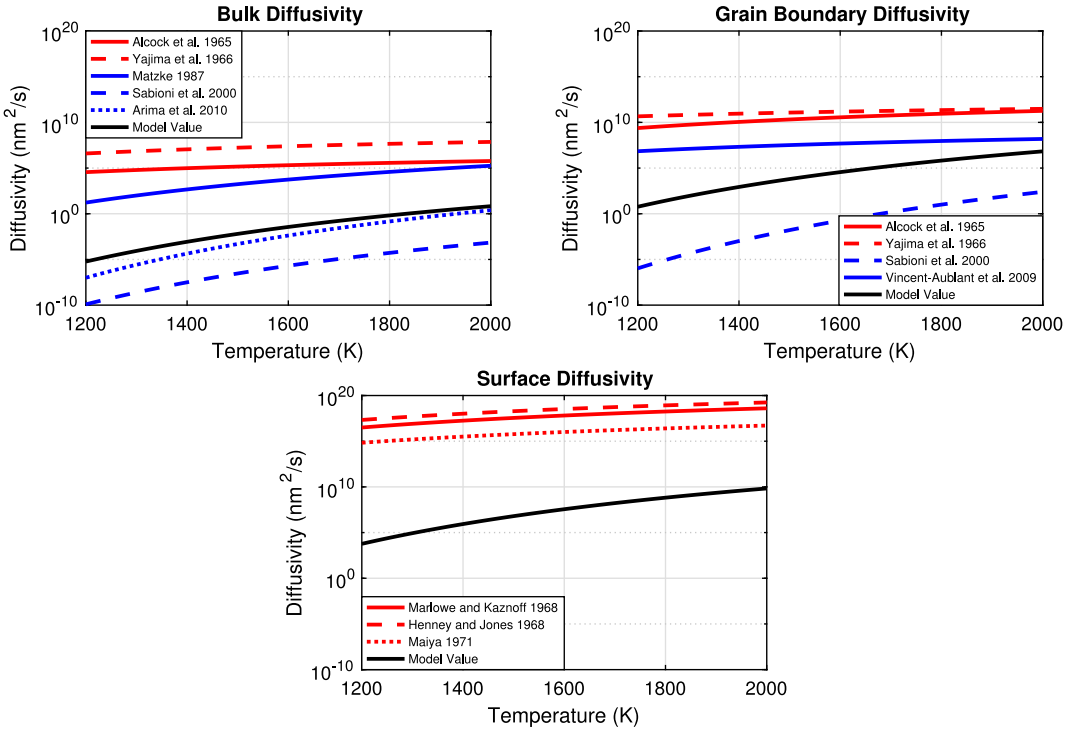


Fig. 3. Various experimental values for bulk [61,62,47,55,56], GB [61,62,55,57], and surface [58–60] diffusivities. Before 1975 there was a systemic error in the methods used to measure diffusivities, causing results to be several orders of magnitude too high [47]. Measurements from before 1975 are shown in red and more recent measurements are shown in blue. The values used in the simulation are shown in black. They fall somewhere near the center of the blue measurements and about 10 orders of magnitude below the red measurements.

void phase equilibrium concentration is always near its equilibrium value, which is set to $c_v^{eq} = 1$ [63].

The value of the interface width ℓ and the temperature T vary between simulations. ℓ is a modeling parameter that is set depending on the size of the domain being modeled. It is selected to be as small as possible but still allow simulations to complete in a reasonable time (three to five elements should cover the distance ℓ). The value of T depends on the conditions being modeled.

To determine the switching value for the S/GB interface ϕ_0, an investigation was carried out to determine the relationship between the error and the value of ϕ_0. To do this, a set of 1D simulations was used with a single surface interface using $\ell = 50$ nm, $T = 1700$ K, a domain size of 200 nm, and 400 elements across the domain. The initial condition put the interface in the center of the domain with a cosine-profile interface. The simulations ran until they reached steady-state conditions. The surface energy was determined by numerically integrating the local free energy density throughout the domain. First, a simulation using constant values was used to verify that the integrated surface energy matched σ_S for the condition $\epsilon = \epsilon_S$ and $\kappa = \kappa_S$. Then simulations were performed using Eq. (13) and values of ϕ_0 ranging from 0.1 to 0.6. The relative errors were calculated based on the difference between the value of σ_S used in Eq. (14) and the value calculated by the integral. The relative memory was determined by taking the ratio of the memory used to the minimum memory from all cases. The relative error and the memory used in the simulations are shown in Fig. 4. The results show that, as expected, smaller values of ϕ_0 produce less error at the expense of increasing the memory usage. The value of $\phi_0 = 0.3$ keeps the error below 5% and the memory increase below 10%. Considering the high uncertainty already in the surface energy value [64], this level of error was deemed acceptable.

With the parameters defined in this section, the model describes the sintering behavior of UO_2. Due to the use of the grand potential formulation, the GB and surface energies can be uniquely defined and are independent of the parameters defining the chemical free energy, such as the vacancy equilibrium concentration in the bulk and on GBs. Also, by using the grand potential formulation from Aagasen et al. [8] spurious phase formations at interfaces are avoided. The model includes accelerated vacancy diffusion at GBs and surfaces, and the diffusion occurs only in the direction parallel to the interfaces. In addition, realistic differences are used for UO_2 between the magnitude of bulk and GB diffusion (six orders of magnitude); surface diffusion is three orders of magnitude larger than GB diffusion (a value that is less than observed experimental behavior but is limited by numerical issues). These differences are much larger than what has been used in previous phase field sintering models. The values from this section will be used in the rest of the work, except where otherwise specified.

3. Model verification

Now that the model is established, it must be verified that it behaves as expected and is implemented correctly. This is done in two ways. First, in Section 3.1 the model is compared with two analytical models to verify that it behaves consistently with the established sintering theory. Then in Section 3.2, the model behavior is tested against the governing mechanisms listed in Section 1. The results of these comparisons are discussed in Section 3.3.

3.1. Comparison to analytical description

In certain idealized test cases, sintering theory allows for problems to be solved analytically and without the use of numerical methods. Two of these problems involve the behavior of two spherical or circular particles as they sinter together. The first of these is the time-evolution of the neck region. The neck width can be expressed in both 2D and 3D cases as [9,7]:

$$\left(\frac{L_{neck}}{D}\right)^n = Kt, \tag{32}$$

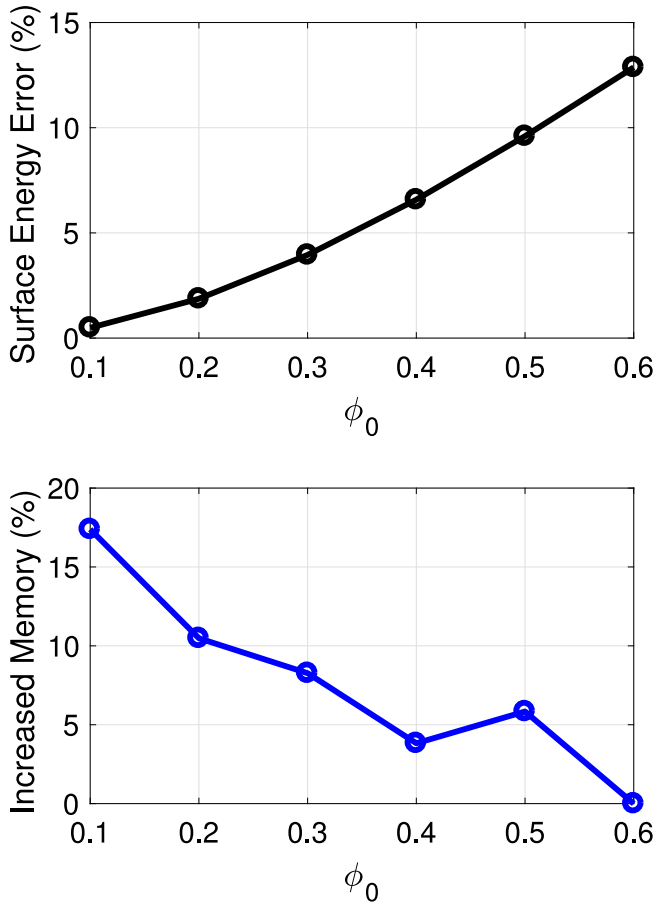


Fig. 4. Selection of the value for ϕ_0 using a 1D surface interface calculation. TOP: Quantification of surface energy error introduced by Eqs. (13), (15) and (17). Small values of ϕ_0 reduce the error. BOTTOM: Quantification of memory usage compared to the simulation which used the least amount of total memory. Small values of ϕ_0 increase memory usage. $\phi_0 = 0.3$ was chosen to keep the error below 5% and the memory increase below 10%.

where L_{neck} is the neck width (first used in Fig. 1), D is the original particle diameter, t is the sintering time, K is a simplified fitting parameter that depends on D, the diffusion rate, surface energy, molar volume, and temperature, and the value of the coefficient n depends on the rate-controlling diffusion process. For GB-controlled diffusion, $n = 6$; for surface-controlled diffusion, $n = 7$ [9].

The second model describes the equilibrium contact angle (dihedral angle) of the sintered material. This is a vital expression both in experiments and simulations [64]. The relationship between dihedral angle and the ratio of GB to surface energy is [65,7]:

$$\frac{\sigma_{GB}}{\sigma_S} = 2\cos\left(\frac{\theta}{2}\right), \tag{33}$$

where θ is the dihedral angle. When Hall et al. reviewed surface energy data [48], they found that not only did the dihedral angle measurements vary significantly, but that the effect of temperature changed depending on the experimental setup. However, two of the major trendlines crossed at 1816 K where the dihedral angle was measured as $\theta = 163°$. This temperature and angle represent the most reliable experimental data on dihedral angles for UO_2.

3.1.1. Simulation descriptions

For both analytical model comparisons, the same basic simulation setup was used. Two 2D particles with a diameter of 300 nm were placed side by side in a 680 nm × 580 nm mesh. Mesh adaptivity was used, with the coarsest mesh having an element size of $(20\text{ nm})^2$ and the finest with an element size of $(5\text{ nm})^2$. The interface width was set to $\ell = 20$ nm and the temperature was set to $T = 1816$ K. The initial condition interface used a hyperbolic tangent profile of thickness ℓ, which is a reasonable approximation to the actual interface profile [66].

Three simulations were performed to test the neck-growth behavior. The first was a GB-controlled diffusion case with the surface diffusion weight from Eq. (24) set to $w_S = 1$ and the second was a surface-controlled case with the GB diffusion weight set to $w_{GB} = 1$. The third case included both effects. The neck width was measured by looking at ϕ along the line $x = 340$ nm and measuring the distance between the two locations where $\phi = 0.5$. If the points fell between two mesh elements, then the exact position was estimated using linear interpolation. The neck width evolution was fit to a power law $L_{neck} = at^b$ and solved for K and n according to $n = \frac{1}{b}$ and $K = \left(\frac{a}{D}\right)^n$.

Three cases were likewise compared to test the effects of interface energies on the dihedral angle. The value of σ_S was changed according to $\sigma_S = \sigma_{GB}$, $\sigma_S = 2\sigma_{GB}$, and $\sigma_S = 3\sigma_{GB}$. For each case, the simulation ran until $t = 50$ s and the angle was measured by hand along the line $\phi = 0.5$.

3.1.2. Simulation results

An example of the simulation evolution is given in Fig. 5. Also shown are a white line along the contour of $\phi = 0.5$ and the measured dihedral angle.

Fig. 6 shows how the neck width evolved in the three cases discussed in Section 3.1.1. For the GB-controlled case, Eq. (32) was fit to parameters $K = 2.99 \times 10^{-6}$ and $n = 7.42$. For the surface-controlled case, Eq. (32) was fit to parameters $K = 0.037$ and $n = 6.35$. The case with both GB and surface diffusion active behaved identically to the surface-controlled case. The GB-controlled case saw neck growth proceed at a significantly lower rate than the other cases. Considering the large difference between w_S and w_{GB} for UO_2 this is not surprising.

The first two seconds of the GB-controlled case requires further discussion. While the hyperbolic tangent profile used in the initial condition is a reasonable approximation, there is always an adjustment period at the beginning of a simulation. In many cases this has a negligible effect on the simulation results. But in the case of the GB-controlled simulation it caused some perturbation in the neck-width measurements.

Fig. 7 shows the results of the dihedral angle measurements from the simulation compared to the analytical expression in Eq. (33) and the experimental value for the given temperature. The maximum difference in the measured angle and the analytical expression is 15° at $\sigma_S/\sigma_{GB} = 1$. As the ratio increases, the agreement improves. Two factors play into this trend in the error. First, in Section 2.3 it was established that there is an inherit error in the surface energy caused by the surface-GB switching function used for implicit solvers. Theoretically this error should increase as σ_S/σ_{GB} increases. This would cause the trend to be less accurate angle measurements as σ_S/σ_{GB} increases. Another source of error in the angle measurement comes from the fact that the angle is measured by hand on a diffuse interface. As the angle sharpens the measurements become more difficult, causing the error to increase. This matches the trend seen in the results. It is also worth noting that the model predicted an angle of 156° for $\sigma_S/\sigma_{GB} = 2$, the energy ratio for UO_2, which is in good agreement with the measured angle for UO_2 of 163° [48].

3.2. Verification of thermodynamics and kinetics

In Section 1, it was established that two mechanisms which must be captured in sintering are the sintering stress and accelerated diffusion on GBs and surfaces. In this section, these behaviors are verified. The ability of the phase field model to accurately represent grain coarsening is well established and will not be investigated here [41,8]. The sintering stress is primarily controlled by the surface and GB energies and

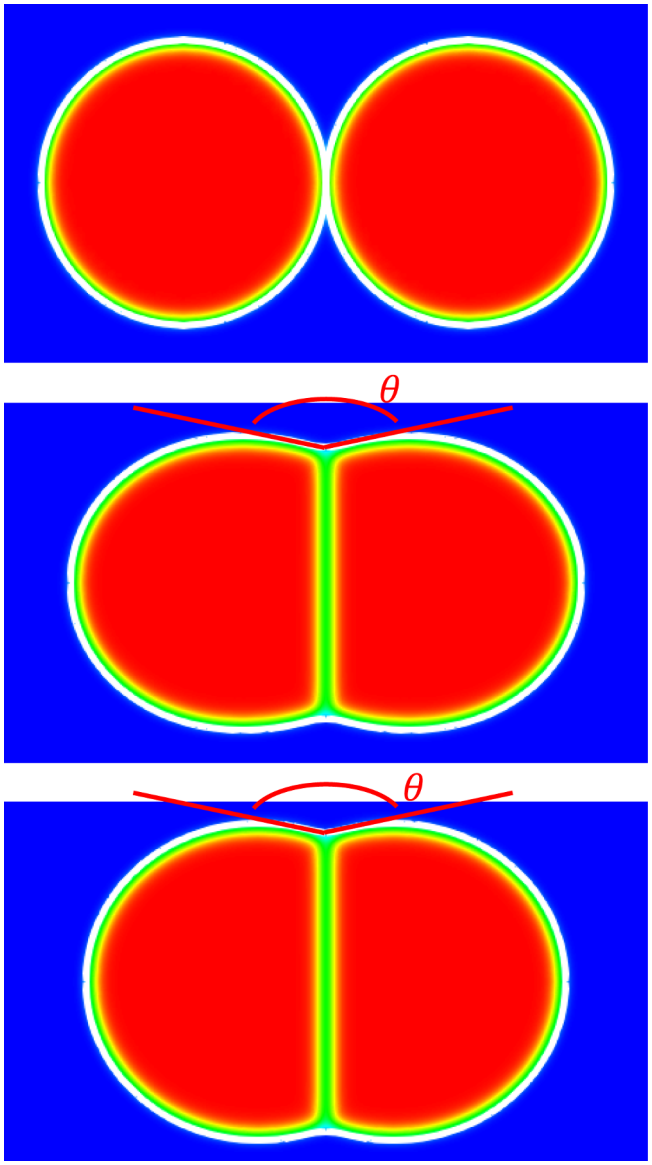


Fig. 5. Demonstration of the two-particle symmetric simulation results. Coloring is according to the variable $\lambda = \sum \eta_i^2$, with a white line showing the position of $\phi = 0.5$. Based on the ϕ contour, the dihedral angle and neck width were measured. The dihedral angle stayed consistent once it reached equilibrium, even as the neck continued to grow. All images are taken from the case using the material parameters for UO_2, with similar microstructure evolution shown in the other cases. TOP: 0 s, MIDDLE: 1.8 s, BOTTOM: 3.5 s.

non-uniform diffusion is controlled through the surface and GB diffusion weights. The effects of changing both of these factors will be quantified to determine if they change the sintering rate as expected.

This section also quantifies the difference these effects have in 2D and 3D simulations. 3D simulations are more accurate to real-world geometry, but 2D simulations have been commonly used in sintering models because they are computationally cheaper [18,20,7,67,68,30]. The effects of dimensionality are quantified by replicating each simulation in both 2D and 3D and comparing the results.

3.2.1. Simulation descriptions

The first set of simulations tested the effects of changing the surface to GB energy ratio, σ_S/σ_{GB}. The difference between these two energies is the driving force for sintering, and therefore should affect the densification rate. The 2D simulations were set up with four circular particles with a uniform diameter of 300 nm arranged in a square formation. This

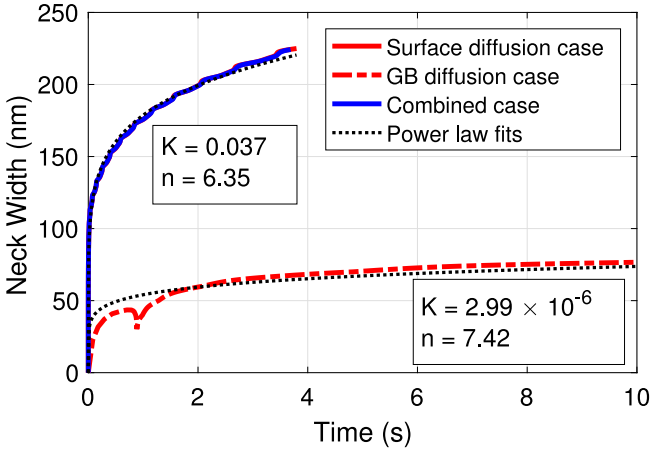


Fig. 6. Evolution of the neck width with time for the surface diffusion case ($\sigma_{GB} = 1$), the GB diffusion case ($\sigma_S = 1$), and the combined case. Power law fits are also included. In the combined case, surface diffusion dominates, causing the behavior to almost perfectly match the surface-diffusion case. The GB diffusion case progressed much slower. In the GB diffusion case, adjustment of the interface from the initial condition near the beginning of the simulation had some impact on the measured neck width, causing a dip early in the plot.

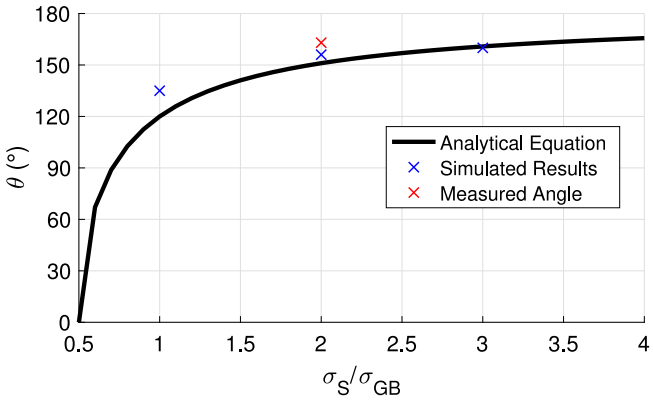


Fig. 7. Dihedral angle measurements compared to the analytical expression which is a function of the surface to GB energy ratio (σ_S/σ_{GB}) and experimental measurement of dihedral angle. The simulated angle shows reasonable agreement with both the analytical expression and the experimental measurement for this temperature.

created a closed pore in the center of the domain. The mesh was $(680\text{ nm})^2$ with $\ell = 40$ nm and $T = 1600$ K. Mesh adaptivity was used with elements ranging from $(10\text{ nm})^2$ to $(40\text{ nm})^2$. The simulation time was 400 s.

The 3D simulations used the same conditions described for the 2D simulations with some additions. Eight spherical particles were arranged in a cube formation. This creates an open pore, unlike the closed pore in the 2D simulations. The height of the mesh was 300 nm such that only half of each particle was inside the mesh. Zero-flux boundary conditions ensured that the top and bottom planes remained symmetry planes. This arrangement was chosen for three reasons:

- A smaller mesh reduces the computational cost
- The 2D simulation setup is equivalent to four 3D cylinders of infinite length. This 3D symmetric setup is also infinite in length.
- A direct comparison of the area of the closed pore in 2D and the volume of the open pore in 3D is not possible. Therefore, rather than measure the total pore volume, the 3D simulation measures the pore cross-sectional area along the top plane of symmetry. This area is equal to the 2D pore area at the initial condition, which allows for the areas to be determined in the same manner for direct

comparison. The cross-sectional area will be referred to as the pore area to simplify discussion of the results.

Several different values of σ_S were used while σ_{GB} was held constant. A total of six simulations were performed. A pair (2D and 3D) of simulations with $\sigma_S/\sigma_{GB} = 1$, a pair with $\sigma_S/\sigma GB = 2$ (correct value for UO_2), and a pair with $\sigma_S/\sigma_{GB} = 3$ were carried out.

The second set of simulations used the same setup as the first set. However, rather than test the thermodynamic driving force, they tested the kinetic effects of the surface and GB diffusivities. Four additional simulation pairs were performed. In one 2D/ 3D pair the surface diffusivity weight was set to $w_S = 1$ in order to isolate the effects of GB diffusivity. In the other pair the GB diffusivity was similarly set to $w_{GB} = 1$ to isolate surface diffusivity. These were then compared to the pair with both surface and GB effects, which is the second pair from the previous set of simulations. The GB diffusion-only case is not expected to sinter in either 2D or 3D. The shape evolution of particles relies on surface diffusion, and since this is retarded, the particles should not be able to pack together any tighter than their initial packing. The surface diffusion-only case is expected to sinter in 3D, but not in 2D. Vacancies from closed pores migrate primarily via GB diffusion. Without accelerated GB diffusion, the closed 2D pore would not be able to evacuate its vacancies. But the 3D open pore vacancies will still have a surface path that allows the pore to close.

The third set of simulations primarily examined the effect of geometry on grain coarsening during sintering. The 2D simulation had two non-uniform circular particles with diameters of 300 nm and 200 nm placed horizontally in a $(580\,\text{nm})^2$ mesh. The interface width was $\ell = 40$ nm and the temperature was $T = 1816$ K. The 3D simulation was based on the 2D setup with the additions that the particles were hemispheres which used the plane of symmetry to reduce computational cost and the mesh height was 300 nm. The simulation terminated once the order parameter for the smaller particle had a maximum value below 0.5. The GB position was determined from the x-coordinate where $\eta_1 = \eta_2 \approx 0.5$ along the line $y = 290$, $z = 0$.

3.2.2. Simulation results

Fig. 8 shows an example of microstructure evolution of the four-particle 2D simulations and the eight-particle 3D simulations used to quantify the effects of thermodynamics and kinetics. White circles in the center images show the pore areas that were measured in each simulation.

Fig. 9 shows plots of the pore area with time for the different energy ratios. The linear region of each curve was fit to a linear function of the form $A = p_1 t + p_2$ where A is the pore area and t is the simulation time. The fits are also shown in Fig. 9. The slopes are plotted vs energy ratio in Fig. 10. The slope magnitude increased as the energy ratio increased. This is the expected result and shows that the sintering stress is acting as a thermodynamic driving force for sintering.

The results for the set of simulations that tested the effects of diffusivity are given in Fig. 11. The red lines correspond to the black lines in Fig. 9. The 3D surface-controlled case exactly matches the 3D combined case, which suggests that the overall pore closure rate is surface-controlled. This behavior matches Fig. 6 which shows neck growth to be surface controlled. As expected, both the 3D GB-controlled diffusion case and the 2D surface-controlled diffusion case showed no pore closure. The most surprising result is that the 2D GB-controlled case actually increased the pore area. The volume change was too large and long-lived to be explained by interface adjustments. Rather, it appears that the model decreased the overall energy of the system by diffusing vacancies from the exterior void to the interior pore along GBs. The causes of this result will be discussed more in depth in Section 3.3.

The linear fits for these simulations were not calculated because the red line fits are already included in Fig. 9, the black line fits have a slope of 0 in 2D and match the red line in 3D, and the blue line fit for 3D also has a slope of 0. The only line where a slope fit would be yield new information is the one line with the positive slope.

Results for the simulations using asymmetric particles to measure grain coarsening are shown in Fig. 12. The microstructure evolution of the 2D and 3D cases are both shown at several intermediate times and the GB location as a function of time is also included. These results show that grain coarsening occurs much faster in 3D than in 2D. In 3D, the smaller grain has been completely consumed within 9.2 s; while in 2D, it takes 24.3 s. The larger surface area for vacancies to diffuse through in 3D appears to more than compensate for the increased volume that must be transported. However, the GB itself does not move as far in 3D. This appears to be related to the geometry. The radius for a circle with the area of two smaller circles is

$$r = \sqrt{r_1^2 + r_2^2}, \tag{34}$$

and the radius of a sphere with the volume of two smaller spheres is likewise

$$r = \sqrt[3]{r_1^3 + r_2^3}. \tag{35}$$

Based on these equations and the values used in the simulation, one would expect the GB to end near $x = 370$ nm in 2D and $x = 354$ nm in 3D. Each of these is slightly lower than the measured GB position, which is reasonable since the center of the particle is not fixed nor is the resulting particle perfectly spherical.

3.3. Discussion

A total of five sets of simulations have been used to verify that the grand potential model is accurately capturing sintering effects. The model makes reasonable predictions of the neck growth rate and the dihedral angle, compared to analytical models. In addition, the behavior of the sintering stress was consistent with expectations. Interestingly, the rate of the decrease in the pore area varied close to linearly with the GB to surface energy ratio. However, additional data would be needed to draw firm conclusions. The impact of the GB and surface diffusion also behaved as expected for most of the simulations. Also note that the behavior of surface and GB diffusion would be different in larger systems. Both open and closed pores exist in real compacts, such that both GB and surface diffusion are controlling the densification.

The only unexpected result was that a closed-pore 2D simulation of GB-only diffusion showed the void increase in size. Reverse sintering like this is a documented phenomenon [69]. However, the physics necessary for reverse sintering to be observed in experiments, such as high gas pressure and sudden temperature changes, were not present in the simulation. To study this further, the 2D GB-controlled case was simulated again, but allowed to continue until the pore closed. During the simulation, the total free energy of the system was calculated, as were the total bulk, surface, and GB energies, as shown in Fig. 13. Pore growth ceased after about 1000 s and the pore closed after 10,800 s.

Interestingly, Fig. 13 shows that the surface energy decreased even as the pore area was increasing. This indicates that the pore shape change increased the pore area but decreased the amount of surface, and was therefore energetically favorable to the system. Fig. 13 also shows the microstructure at several times during the simulation. Unlike in Fig. 8, here the primary shape change occurs at the GBs, creating a non-equilibrium X-shape when the pore closes. This is reasonable because the GBs are the major source of diffusion for the system and the GBs are the regions where motion occurred. The domain will eventually reach its equilibrium shape, similar to the final shape shown in Fig. 8, after much more time. The GB motion may have helped the initially concave pore to form a convex shape with a larger area but smaller perimeter.

Based on the results of all of the simulations in this section, several conclusions can be drawn. First, the UO_2 parameters clearly exhibit surface diffusion-controlled kinetics. Not only is the surface diffusivity 1,000 times larger than the GB diffusivity, but both sets of simulations that compared surface-only diffusion to surface and GB diffusion showed identical results. Second, 3D effects are vitally important to get realistic

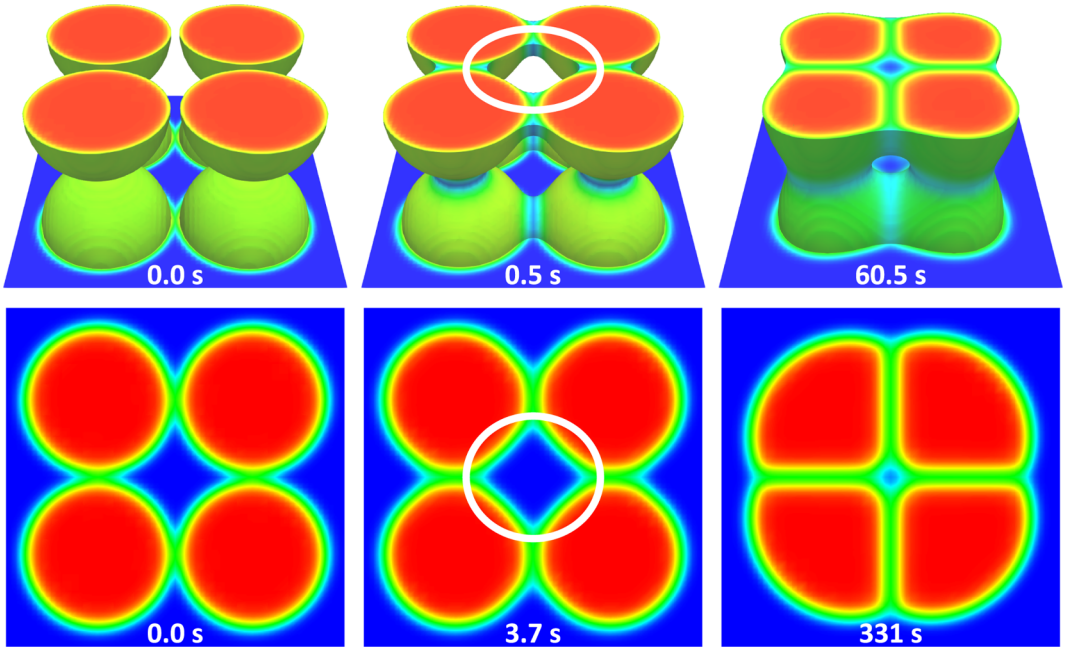


Fig. 8. Demonstration of 2D and 3D pore closure simulations using $\sigma_S/\sigma_{GB} = 2$. TOP: 3D 8-particle simulation of pore closure. The bottom domain's surface is shown along with the region $\phi \leqslant 0.2$. BOTTOM: 2D 4-particle simulation of pore closure. In both cases the coloration represents the variable λ from Eq. (29). From left to right both rows show an initial condition, an intermediate stage, and the point of pore closure. The 2D pore area was compared against the 3D pore cross-section on the top surface of the domain. Both areas are circled in white in the two center images. The time of each image is given. Similar microstructures were produced for the other cases.

results. All of the simulations that compared 2D and 3D behaviors showed 3D progressed significantly faster in terms of pore closure and GB motion. One cause for this is that pores that are closed off from the external surface in a 2D domain may be open to the external surface in a 3D domain. Since surface diffusion is the rate-controlling effect, this means that 2D domains will underestimate the closure rates of these pores Because of these geometric effects, any attempt at validating the model against experimental results will require 3D simulations.

4. Comparison against experiments

All computational models should be validated against experimental data to ensure their accuracy. While macroscale models like the master sintering curve have been well-validated [70], validation of microstructural models is more difficult. The MC Potts model has only been compared to data once [15] and no comparison to data has been carried out for phase field sintering models [6].

There are three challenges when validating microstructural sintering models. First, a method is needed to create the structure of the initial powder compact. In a green body there are millions of individual particles of various sizes arranged in an unorganized lattice, and a method is needed to create a simulated compact with similar characteristics. Second, a method is needed for determining the compact volume that is similar to the approaches used in experiments. Third, the computational cost of representing the large number of particles found in actual green bodies is too high, and therefore the microstructural

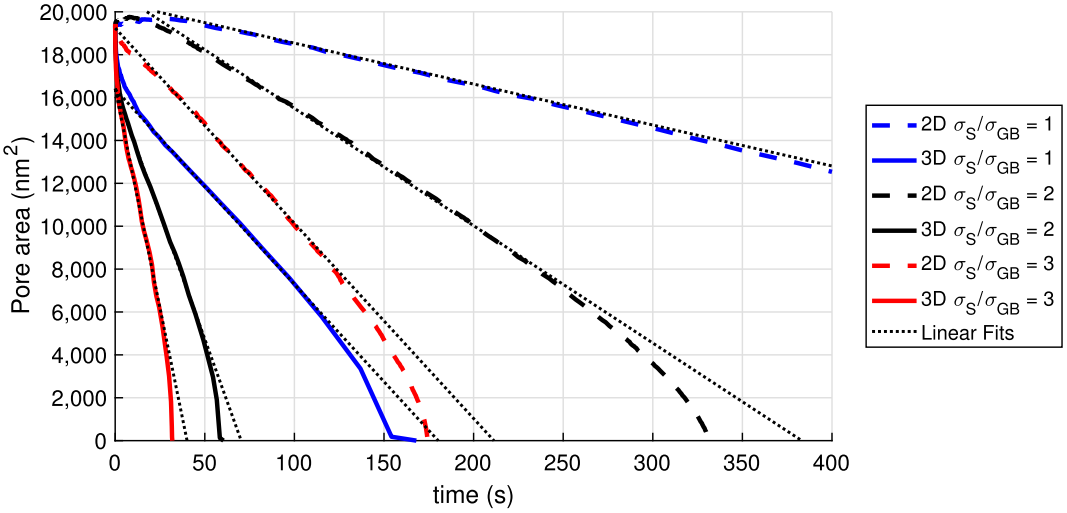


Fig. 9. Pore area measurements for the 2D 4-particle and 3D 8-particle sintering stress simulations. In these simulations the surface to GB energy ratio, σ_S/σ_{GB} was varied to verify that the energy difference is acting as the driving force. The rate of pore closure increases as σ_S/σ_{GB} increases. Also shown are fits to the linear regions of each curve.

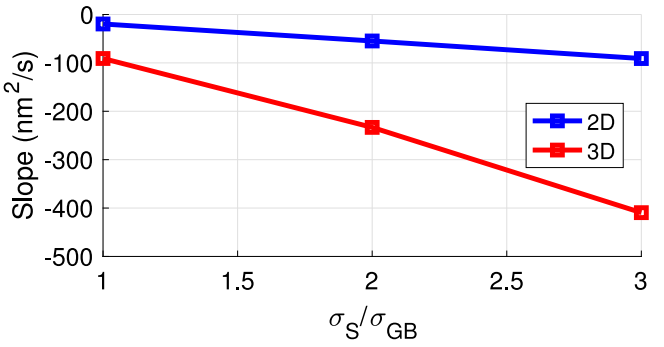


Fig. 10. Slopes of the linear fits in Fig. 9 as a function of the surface to GB energy ratio. The change appears to be linear in 2D and nearly linear in 3D, but more data are needed to verify the trend.

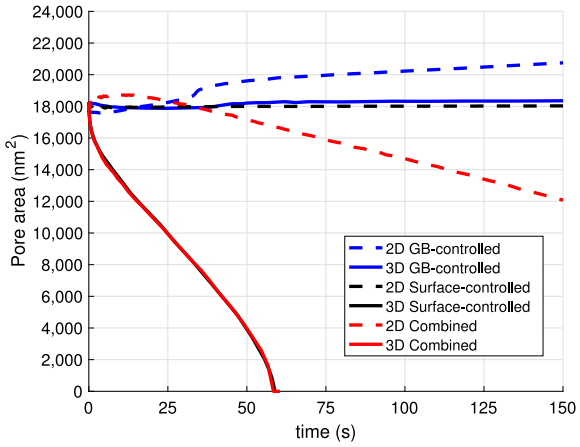


Fig. 11. Results of 3D 8-particle and 2D 4-particle diffusivity simulations. In this set of simulations the surface and GB diffusivities were modified. In one pair of simulations the surface diffusion weight was changed to $w_S = 1$ in order to make it GB-controlled. In another pair the GB diffusion weight was changed to $w_{GB} = 1$ in order to make the process surface-controlled. The third pair used the weights described in Section 2.3 in order to include both effects. The third pair corresponds to the black lines in Fig. 9.

model represents the behavior in a much smaller body. Methods for handling the first two challenges are presented here, and the consequences of the third challenge are discussed.

In this section, an attempt is made to begin validating the grand potential sintering model using experimental data available in the literature. First, the approach taken to create the initial condition is described in Section 4.1 and the approach for measuring the volume in Section 4.2. The simulation is then described in Section 4.3 and the results are presented in Section 4.4. The results are discussed in Section 4.5.

4.1. Initial condition generation

A sintering simulation will not yield realistic results unless the initial condition (IC) meets the following criteria:

- The IC contains a reasonably large number of particles,
- The particles follow a realistic size distribution,
- The particles contact one another to form a single body without any particle overlap,
- The bulk density is approximately the same as real powder compacts,
- The particles do not form an organized lattice.

A particle packing algorithm was developed to generate ICs based on

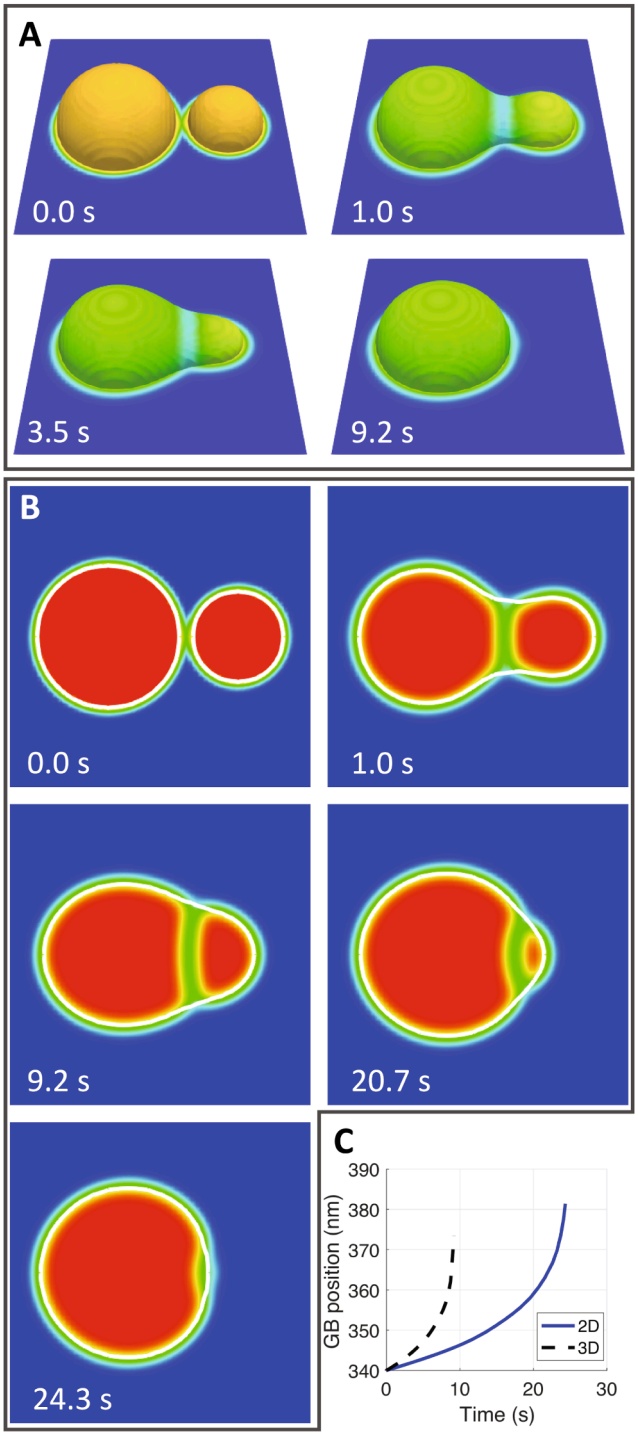


Fig. 12. Results from 2-particle simulations using asymmetric particles. The particle radii are 150 nm and 100 nm with their initial contact point at x = 340 nm. A: 3D simulations with the bottom surface of the domain and the surface $\phi = 0.1$ shown at several timesteps. B: 2D simulation with the line $\phi = 0.1$ shown in white. C: Position of the GB as a function of time. The 3D case exhibits faster overall grain growth even though the GB does not move as far. In A and B the coloring is according to $\lambda = \sum \eta_i^2$.

these criteria. It is demonstrated in Fig. 14. First, spherical particles are randomly sized according to a predetermined size distribution. Then, particles are placed, largest to smallest, in the domain at random positions one at a time. If a particle overlaps with a pre-existing particle, then the new particle is moved to a new location until one is found in which it does not overlap any other particles.

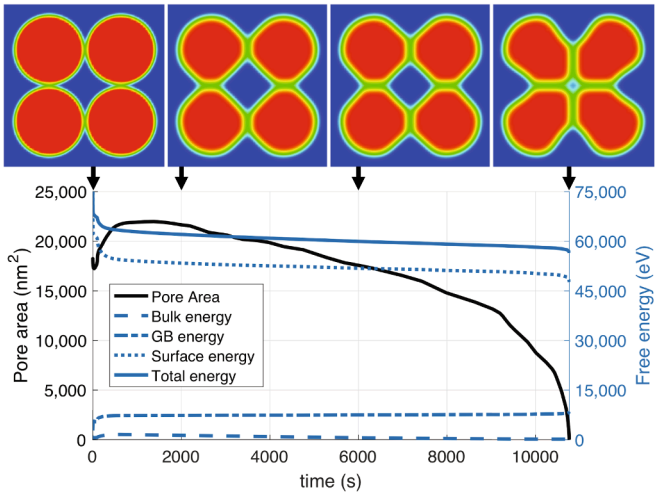


Fig. 13. 2D GB-controlled case from Fig. 11 performed a second time for a longer time period. The black line shows the pore area over time. Pore growth occurs for the first 1000 s before the pore begins to close. The blue lines show the bulk, GB, surface, and total energy of the simulation. The microstructure of the simulation is shown at four times with arrows pointing to their corresponding times on the graph. The pore changes from a concave shape at the initial condition to a convex shape. This causes the pore perimeter to decrease, which reduces the surface energy, while increasing the pore area. The microstructure at the time when the center pore disappears is very different than that shown in Fig. 8. However, the equilibrium geometry would be identical to Fig. 8 but it would take much more time to reach it. This microstructure evolution is unphysical due to the nature of the test being performed.

Once all of the particles are placed, a potential field is applied to the domain. The potential is lowest at the bottom and highest at the top. One at a time, the particles attempt to move a set distance down the potential gradient to lower their potential. If this movement causes the particle to leave the domain or overlap another particle, then a penalty is applied which increases the potential. The particle may then attempt to push other particles out of the way, calculate a new gradient which accounts for the field and the penalty and move again, or return to its original position. Particles continue cycling through this process until none of the particles can move.

Once the cycle is finished, particles that are not in contact with the main body of particles can be manually removed. Finally, the particle positions and radii are output to a text file that can be read by MARMOT and used to generate the simulation's IC.

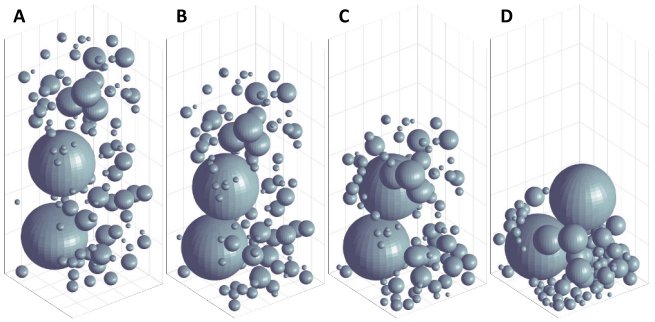


Fig. 14. Demonstration of the algorithm used to generate ICs for sintering simulations. A: The particles are randomly placed such that no two overlap and all are completely inside the domain. B: A potential field pulls all of the particles down, and penalties are applied to prevent overlap and exits from the domain. C: Particles begin to gather at the bottom of the domain. D: The script finishes when no particle is able to move and minimize its potential. Any particles that are not in contact with the larger body (such as the one seen on the bottom right edge of D) are removed.

4.2. Density approximation

In experiments, there are typically two methods used to measure pellet volumes for density calculations [71]. First there is liquid immersion. A pellet is submerged in a liquid such as water and the volume change of the liquid is measured. The other method is to measure the pellet dimensions with a caliper or similar tool and calculate the volume. Both of these methods are difficult to replicate in a microstructural simulation. In the case of liquid immersion, the simplest computational method to replicate this would be a flood algorithm, which colors regions according to the dominant continuous variable [42]. Internal pores could then be separated from external pores based on continuousness. However, the initial condition method described above generates a completely open microstructure. A flood algorithm would classify the interconnected porosity as a single pore. In actual experiments, forces such as surface tension and gas bubbles would prevent complete wetting of the material, but a flood algorithm does not consider these effects.

Replicating a caliper measurement is complicated by the small number of particles, and thus the small length-scale, used in the phase field simulations. These very small green bodies will have a large amount of surface variation, as shown in Fig. 14. Caliper volume measurements will have a certain level of error based on the surface roughness of the pellet. For a typical green body, this error is negligible. However, for the small green bodies used in phase field simulations, this error becomes large. Any method to approximate the density based on the simulated microstructure must mitigate some of the effects of surface roughness.

A method was developed to approximate the density during simulated sintering. In the method, the volumes and centroids of each particle/grain are used to generate a sphere that has an equivalent centroid and volume. A convex hull is drawn around the centroids, eliminating surface roughness from the system. Finally, a MC integration technique is used to estimate the density [72] inside the convex hull. Points inside the hull are randomly sampled and the number of points that fall within the spheres are counted. Points that fall within multiple spheres (since they only approximate the actual structure, and could thus overlap) are counted once for each of the overlapping spheres. This method is demonstrated in Fig. 15 by overlaying an image of the MC integration over an image of the microstructure.

4.3. Simulation description

The simulation was designed to replicate the 1983 experiment described by Burk et al. [71]. This paper was chosen because it included particle size distribution measurements for green pellets, which allows for the most accurate initial condition possible. However, it only included density measurements at the beginning and end of sintering. It did not include other important measurements such as intermediate density measurements or grain size. Green pellets of stoichiometric UO_2 were heated at a rate of 4 K/min to 1973 K, held at this temperature for two hours, then cooled at a rate of 5 K/min back to room temperature. The particle size distribution was estimated based on the bimodal lognormal distribution from the paper [71]. The initial density was approximately 5.5 g/cm^3 (50.1%) and the final density was approximately 10.7 g/cm^3 (97.5%).

An IC was generated as described above using 120 particles randomly sampled from the distribution. Seven particles had to be removed because they were not in contact with the larger body of particles, resulting in a group of 113 particles inside a 5.52 μm by 5.52 μm by 6.48 μm domain. The interface width was set to $\ell = 120$ nm and a minimum diameter for the initial particles was set to 200 nm. Only 24 solid order parameters were used to represent the 113 particles to reduce computational cost, and the Grain Tracker algorithm in MARMOT [42] was used to avoid coalescence of particles.

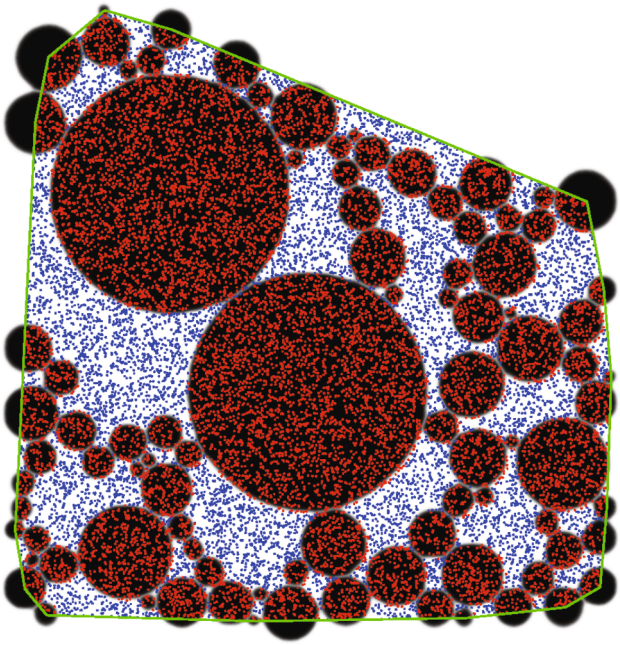


Fig. 15. Demonstration of the MC density estimation technique created by overlaying an image of a convex hull and MC points over an image of the microstructure. A convex hull is wrapped around the grain centroids. Black regions are grains and white regions are voids. The hull is shown in green. Points inside the hull are randomly sampled. Points that fall within grains are marked in red and points in voids are marked as blue. The relative density is calculated by dividing the number of red points by the total number of points.

4.4. Simulation results

The evolution of the 113 particles was simulated using the same temperature conditions as the experiment, and all particles were maintained at the same temperature. The microstructure evolution of the simulation is shown in Fig. 16. The initial density was an exact match with the data. The particles coalesced and sintered as expected. However, there are several particles that break from the main body. The smallest ones then shrink via mass diffusion through the void space, which is unphysical. This breakage suggests that the initial condition was a poor quality powder compact. In the initial compact, the small particles tended to be at the bottom of the simulation domain while larger particles tended to be near the top. This unequal distribution may help account for the poor compact quality.

The estimated density along with experimental data and the temperature profile are given in Fig. 17. All pores were eliminated before reaching the sintering temperature and the system spent the second half of the simulation at 100% theoretical density. The final relative density from the experiment was 97.5%.

4.5. Discussion

The MC density approximation worked as intended, but it appears that the IC needs additional development. Some particles separated from the main body, and although the initial density was reasonable, the particles self-sorted by size. The mechanism of self-sorting is readily apparent. Small particles can travel through the spaces between the large particles, which allows them to reach the bottom of the domain where the potential field is lowest. However, self-sorting by size is not observed in experimental powder compact. It is possible that more mixing is necessary in the initial condition to more accurately capture the sintered microstructure.

The sintering rate of the simulation was physically reasonable, though the simulated body reached 100% relative density while the experiments did not. In addition, Fig. 17 shows that the body reached

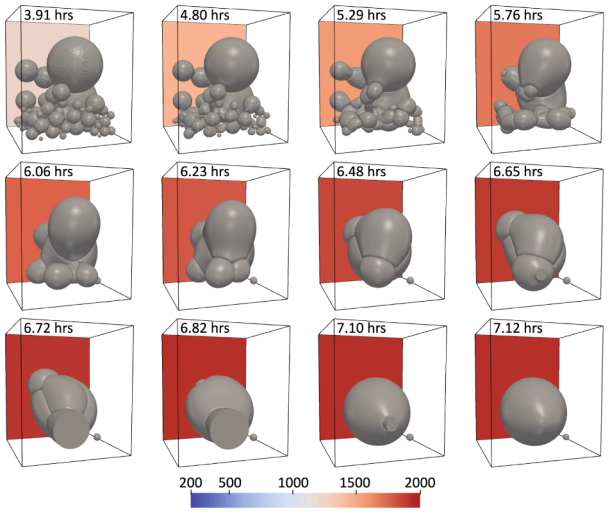


Fig. 16. Evolution of the 113 particle sintering simulation. The microstructure is shown in gray. The back edge of the simulation domain is shaded by temperature. Some particles broke off from the main body, signifying the initial condition may have been a poor compact.

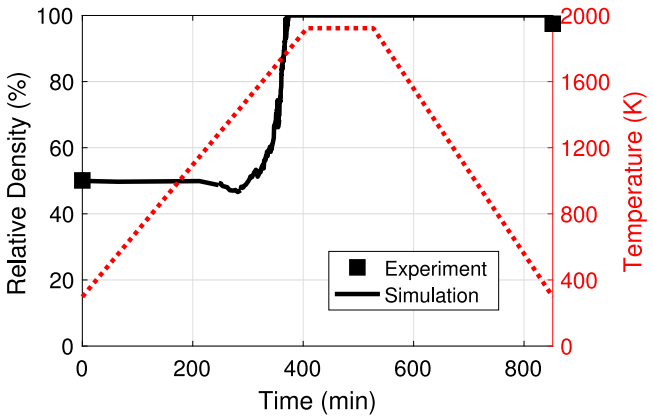


Fig. 17. Relative density during the 113 particle sintering simulation, where the density was approximated using the method discussed in Section 4.2. The measured initial and final relative densities and the temperature profile from Ref. [71] are included for reference. The simulation had the correct initial density but the simulated body reached 100% density, while the experimental body only reached 97.5% density.

100% density quickly, even before the end of the temperature ramp. One likely reason for this rapid densification is the small number of particles used in the simulation. Fewer particles means a higher fraction of open pores, which densify faster than closed pores, as well as shorter distances for vacancies to diffuse before reaching external surfaces, which also speeds up the kinetics. Using more particles for future simulations would result in a larger simulation domain and slower densification, but also higher computational cost. However, in this simulation, the small particles that resulted from the smaller side of the bimodal distribution combined with larger grains quickly. By the time that significant densification started, at around 300 min, all grains with initial sizes less than 400 nm (about half of all the initial particles) had already been consumed. Thus, future simulations could sample particle sizes only from the larger half of the distribution, resulting in a larger domain with the same number of particles. An alternative approach to validating the model is to compare against data collected using small particle compacts, as done by Tikare et al. [15]. They compared against data from copper sintering of cylindrical green bodies with diameters of 40 to 60 μm that were characterized using X-ray synchrotron irradiation.

Another likely reason for the rapid densification is the large uncertainty in the measured property values that inform the diffusivity (see Fig. 3). The values used in this work may not have accurately described the diffusion behavior occurring in the experiments. Uncertainty quantification needs to be carried out to quantify the impact of this property uncertainty on the density change predicted by the model.

The results from the new grand potential sintering model are promising. They show that the model does not experience the under-prediction of the densification rate found in previous phase field models without the Rigid Body Motion model. The results also indicate where modifications can be made to improve future predictions, mainly in increasing the size of the domain modeled and understanding the impact of the large parameter uncertainty. Once this improvements are made, future comparisons to data are needed to fully validate the model.

5. Conclusions and future work

A quantitative grand potential sintering model has been presented that captures the four driving mechanisms of sintering: sintering stress, vacancy/GB interaction, rapid diffusion on GBs and surfaces, and grain coarsening. Small test simulations were used to verify that these mechanisms are accurately captured in the model. Additional test simulations compared the model's behavior with analytical relations derived from sintering theory. In these tests, the model performed reasonably well and the results indicated that densification and coarsening occurs faster in 3D simulations than in 2D.

Comparisons between 3D simulations and experimental data were also carried out. First, the approaches taken for generating the IC and for quantifying the density were presented. However, the IC appears to have resulted in a poor initial compact, as some particles detached from the densifying body. The sintering rate was reasonable, demonstrating that the grand potential model can capture the sintering rate without requiring the Rigid Body Motion model. However, while the simulated final body reached 100% relative density, the experiment only reached 98% density. In addition, the densification occurred too quickly. The cause of this is unclear, though it is likely related to the small size of the green body used in the simulation and the large uncertainty in the measured diffusion properties.

These results fall short of validating the model, but do offer guidance on where improvements could be made in future attempts to validate this model and other microstructure sintering models. The size of the green body could be increased without increasing the computational cost by neglecting small particles, which disappear quickly. Comparisons could also be made to data collected on green bodies that are the same size represented in the simulations. To reduce the property uncertainty, future experimental work and atomistic simulations should seek to determine these values with greater accuracy. Finally, additional work should be done to develop an IC that uniformly distributes the particles by size and better represents the actual particle size distribution.

Data availability

All simulation data produced in this work will be provided upon request. The MARMOT tool is export controlled with access granted on an individual basis. To apply for MARMOT access seehttps://moose.inl.gov/marmot/SitePages/Access.aspx.

CRediT authorship contribution statement

Ian Greenquist: Conceptualization, Software, Validation, Writing - original draft. **Michael R. Tonks:** Funding acquisition, Conceptualization, Supervision, Writing - review & editing. **Larry K. Aagesen:** Resources, Software, Writing - review & editing. **Yongfeng Zhang:** Funding acquisition, Supervision, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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